

PROGRAMMING ASSIGNMENT - 4

Complete Diagnostic Bug Analysis Report (Files 1.c to 5.c)

Problem 1 (1.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
1	10	Missing semicolon after declaration of department.	Syntax Error	Change to <code>int department = 1;</code>
2	12	String literal "90" assigned to integer variable 'marks'.	Semantic / Type Error	Change to <code>int marks = 90;</code>
3	26	Assignment operator '=' used instead of equality comparison '==' inside if statement.	Logical Error	Change to <code>if(x == 10)</code>
4	31	Variable 'score' used without being declared.	Semantic Error	Declare <code>int score;</code> before use.
5	51	Function 'add' called but not declared or prototyped previously.	Compilation Error	Add <code>int add(int, int);</code> prototype.
6	58	Variable 'temp' accessed outside its block scope.	Scope Error	Declare 'temp' outside the inner block.
7	62	Array index 10 out of bounds for 'arr[5]'. [Note: The error description in the image incorrectly states 'arr[5]' while the correction shows 'arr[4]']	Runtime Error	Change to <code>arr[4] = 100;</code>
8	66	Format specifier '%d' used to print a float variable 'salary'.	Format Mismatch	Change to <code>printf("%f ", salary);</code>
9	86	Variable 'result' accessed outside the for loop body block scope where it was defined.	Scope Error	Declare 'result' before the for loop.
10	93	Both 'if' and 'else' branches execute identical code.	Logical Error	Provide proper handling for the else case.
11	97	Function 'displayData' called with no arguments, but implementation expects an integer.	Function Argument Error	Call as <code>displayData(value);</code>
12	133	Semicolon immediately after 'while(z > 0)' creates an infinite empty loop.	Logical Error	Remove the trailing semicolon.
13	138	String literal "A" assigned to single char variable 'grade'.	Type Mismatch	Change to <code>char grade = 'A';</code>
14	142	Format specifier '%s' used to print integer variable 'number'.	Runtime Error	Change to <code>printf("%d ", number);</code>
15	146	Dereferencing a NULL pointer 'ptr'.	Runtime Error	Assign a valid address to ptr before dereferencing.
16	150	Function 'calculateAverage' called but missing a definition or prototype.	Linker Error	Provide declaration and definition.

Problem 2 (2.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
17	10	String literal "90" assigned to integer variable 'marks'.	Type Mismatch	Change to <code>int marks = 90;</code>
18	16	Missing semicolon after printf statement.	Syntax Error	Add a semicolon at the end of the line.
19	24	External variable 'total' referenced but not defined anywhere in the program.	Linker Error	Define <code>int total;</code> globally or link properly.

S.No.	Line No.	Error Description	Error Type	Error Correction
20	42	Attempting to get address of a register variable '&number'.	Syntax Error	Remove register specifier or avoid using '&'.
21	44	String literal "50000" assigned to integer variable 'salary'.	Type Mismatch	Change to <code>static int salary = 50000;</code>
22	51	Variable 'department' accessed outside its block scope.	Scope Error	Declare variable 'department' outside the block scope.
23	61	Variable 'temp' accessed outside the for loop body block scope where it was defined.	Scope Error	Declare 'temp' before the for loop.
24	65	Array index 10 out of bounds for 'arr[5]'.	Runtime Error	Use a valid index from 0 to 4.
25	69	Division by zero error (a/b where b=0).	Runtime Error	Ensure 'b' is not equal to 0 before division.
26	75	Assignment operator '=' used instead of equality comparison '==' inside if statement.	Logical Error	Change to <code>if(value == 100)</code>
27	81	Both 'if' and 'else' branches execute identical print statements.	Logical Error	Modify the else branch logic.
28	95	String literal "A" assigned to single char variable 'grade'.	Type Mismatch	Change to <code>char grade = 'A';</code>
29	117	Variable 'data' used without being initialized (prints garbage value).	Runtime Warning	Initialize <code>int data = 0;</code>
30	121	Dereferencing a NULL pointer 'ptr'.	Runtime Error	Assign a valid memory address to ptr before use.
31	123	External variable 'companyCode' referenced but not defined.	Linker Error	Define companyCode in an external or current module.
32	127	Function 'calculateAverage' called but not declared or defined anywhere.	Linker Error	Provide declaration and implementation.
33	131	String literal assigned to integer variable 'employeeId'.	Type Mismatch	Change employeeId to a string array or change assignment.
34	135	Semicolon after while condition creates an infinite empty loop trap.	Logical Error	Remove trailing semicolon after the while loop header.

Problem 3 (3.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
35	23	Missing semicolon after variable choice declaration.	Syntax Error	Change to <code>int choice;</code>
36	53	Missing colon separator after default keyword inside switch block.	Syntax Error	Change to <code>default:</code>
37	79	Database Full condition checks if index > 5 after incrementing, but index 5 already overflows array boundary of size 5.	Array Out of Bounds Error	Change condition to <code>if(index >= 5)</code> and place before insertion.
38	88	Loop fencepost boundary condition <code>i <= totalStudents</code> accesses uninitialized out of bounds array item.	Logical / Out of Bounds Error	Change to <code>i < totalStudents</code>
39	113	Hardcoded division by zero <code>return total / 0;</code> which crashes the OS process at runtime.	Runtime Error	Change to <code>return total / totalStudents;</code>
40	141	Typo in variable name <code>totalStudent</code> inside printf (defined as totalStudents).	Syntax Error	Change to <code>totalStudents</code>
41	143		Linker Error	

S.No.	Line No.	Error Description	Error Type	Error Correction
		Functions <code>`printGrade()`</code> and <code>`calculateRank()`</code> are called but have no definition.		Implement the referenced functions.
42	152	String literal "101" assigned to scalar integer variable <code>`s.rollNo`</code> .	Type Mismatch	Change to <code>s.rollNo = 101;</code>
43	158	String literal "90" assigned to float variable <code>`s.marks`</code> .	Type Mismatch	Change to <code>s.marks = 90.0;</code>
44	171	Using variable <code>`roll`</code> directly as an index without verification lookup from input value.	Logical Error	Implement lookup matching roll numbers.
45	180	A struct element <code>`students[roll]`</code> cannot be initialized as a <code>`NULL`</code> pointer pointer literal.	Semantic Error	Clear the struct members or use a flag.
46	184	Function <code>`searchStudent()`</code> has no body definition block.	Syntax Error	Add body definition to function.

Problem 4 (4.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
47	104	Loop boundary test expression <code>`i <= totalAccounts`</code> triggers a fencepost overflow error.	Logical Error	Change to <code>i < totalAccounts</code>
48	141	The expression logic <code>`balance > amount`</code> incorrectly alerts "Insufficient Balance" when funds are sufficient.	Logical Error	Change conditional to <code>balance < amount</code>
49	185	Typo in parameter name <code>`totalAccount`</code> inside <code>printf</code> (defined as <code>totalAccounts</code>).	Syntax Error	Change to <code>totalAccounts</code>
50	196	String literal "1001" assigned directly to scalar integer structure component <code>`a.accNo`</code> .	Type Mismatch	Change to <code>a.accNo = 1001;</code>
51	202	String literal "50000" assigned to float field component variable <code>`a.balance`</code> .	Type Mismatch	Change to <code>a.balance = 50000.0;</code>
52	224	Structure block element component assignment <code>`accounts[accNo] = NULL;`</code> is completely invalid.	Semantic Error	Reset fields to zero or toggle a valid/active flag.
53	237	Comparing string content addresses directly via comparison operator <code>`==`</code> instead of <code>strcmp</code> .	Semantic Error	Change to <code>if(strcmp(name, accounts[0].name) == 0)</code>
54	248	Function <code>`generateAverage()`</code> called in statistics but has no definition.	Linker Error	Implement the referenced function.
55	259	File opened with read permission "r" but immediately calls write parameter <code>`fprintf`</code> .	Runtime Error	Change file opening permission to "w".

Problem 5 (5.c)

S.No.	Line No.	Error Description	Error Type	Error Correction
56	25	Missing semicolon termination after variable choice declaration statement.	Syntax Error	Change to <code>int choice;</code>
57	63	Missing colon character indicator after default keyword statement block setup.	Syntax Error	Change to <code>default:</code>
58	93	Library Full safety limit verification checks <code>> 20</code> after <code>totalBooks</code> was already written to index 20 (out of bounds).	Array Out of Bounds Error	Change to <code>if(totalBooks >= 20)</code> and place before insertion.
59	104		Logical Error	Change to <code>i < totalBooks</code>

S.No.	Line No.	Error Description	Error Type	Error Correction
		Loop fencepost error tracking condition <code>`i <= totalBooks`</code> triggers uninitialized memory scanning.		
60	159	Typo in variable name reference <code>`totalBook`</code> inside <code>printf</code> block.	Syntax Error	Change to <code>totalBooks</code>
61	163	Function links <code>`calculateFine()`</code> called inside report but missing implementation definition.	Linker Error	Implement the referenced function.
62	170	String literal pointer assigned directly to scalar integer element component <code>`b.bookId`</code> .	Type Mismatch	Change to <code>b.bookId = 101;</code>
63	178	String literal literal forced inside scalar float element variable component <code>`b.price`</code> .	Type Mismatch	Change to <code>b.price = 500.0;</code>
64	191	Comparing string content memory location pointers via <code>`==`</code> instead of standard <code>`strcmp()`</code> .	Semantic Error	Change to <code>if(strcmp(title, books[0].title) == 0)</code>
65	202	Compound object configuration element value assigned to a pointer literal <code>`NULL`</code> .	Semantic Error	Clear the struct members individually or set an active flag.
66	216	Function <code>`calculateAveragePrice()`</code> called but missing definition.	Linker Error	Implement the function body.
67	227	File opened with read mode <code>"r"</code> but immediately updates contents using <code>`fprintf`</code> .	Runtime Error	Change file opening permission to <code>"w"</code> .
68	258	Allocates only 5 bytes of heap space via <code>`malloc(5)`</code> but loop writes 20 integer items (80 bytes), causing a severe heap overrun.	Runtime / Memory Error	Change to <code>malloc(20 * sizeof(int))</code>
69	265	Double free error: calling <code>`free(ptr)`</code> twice on the same memory reference pointer.	Runtime Error	Remove the redundant secondary <code>free(ptr);</code> block line.